

[Click here to replace this description.](#)

– 1 pt (b) setup

- 1 pt (b) formula applied correctly

- 1 pt (b) answer 3/4

– 1 pt (c) setup

- 1 pt (c) formula applied correctly

- 1 pt (c) 242/243

- 9 pts Incorrect

– 1 pt Use value of c

Question 2

Problem 2

2 / 2 pts

✓ - 0 pts Correct

- 2 pts incorrect

– 1 pt [Click here to replace this description.](#)

– 0 pts point taken but note that here we would have just left n as $\log_{0.94}(0.5)$ or we would have rounded to 11 since that gets you at least 50%

– 3 pts major errors in problem

Question 5

Problem 5

5 / 5 pts

✓ – 0 pts Correct

– 2 pts (a) incorrect pmf (asking for max of dice outcome, not sum)

– 1 pt (b) incorrect cdf

– 0.5 pts should use step function when graphing cdf

– 1 pt didn't show cdf function values

– 5 pts cannot find problem 5 solution

– 1 pt cannot find cdf graph

– 2 pts incorrect pmf

– 1 pt (a) incorrect pmf notation (should be over 36)

independent.

Solving
this

$P(\text{at least one hit}) = 1 - P(\text{No hits in } n \text{ throws})$
• Prob. of no hits in n throws: $P(\text{no hits}) = 0.94^n$

$$\rightarrow 1 - 0.94^n = 0.5 \rightarrow 0.94^n = 0.5 \rightarrow \ln(0.94^n) = \ln(0.5)$$

$$n \ln(0.94) = \ln(0.5) \rightarrow n = \frac{\ln(0.5)}{\ln(0.94)} = \frac{-0.6931}{-0.06188} = 11.2 \rightarrow \boxed{12}$$

Round up

$(X=1)$: Only one outcome $\Rightarrow (1,1) \rightarrow \frac{1}{36}$

$P(X=2)$: outcomes $\Rightarrow (1,2), (2,1), (2,2) \rightarrow \frac{3}{36} = \frac{1}{12}$

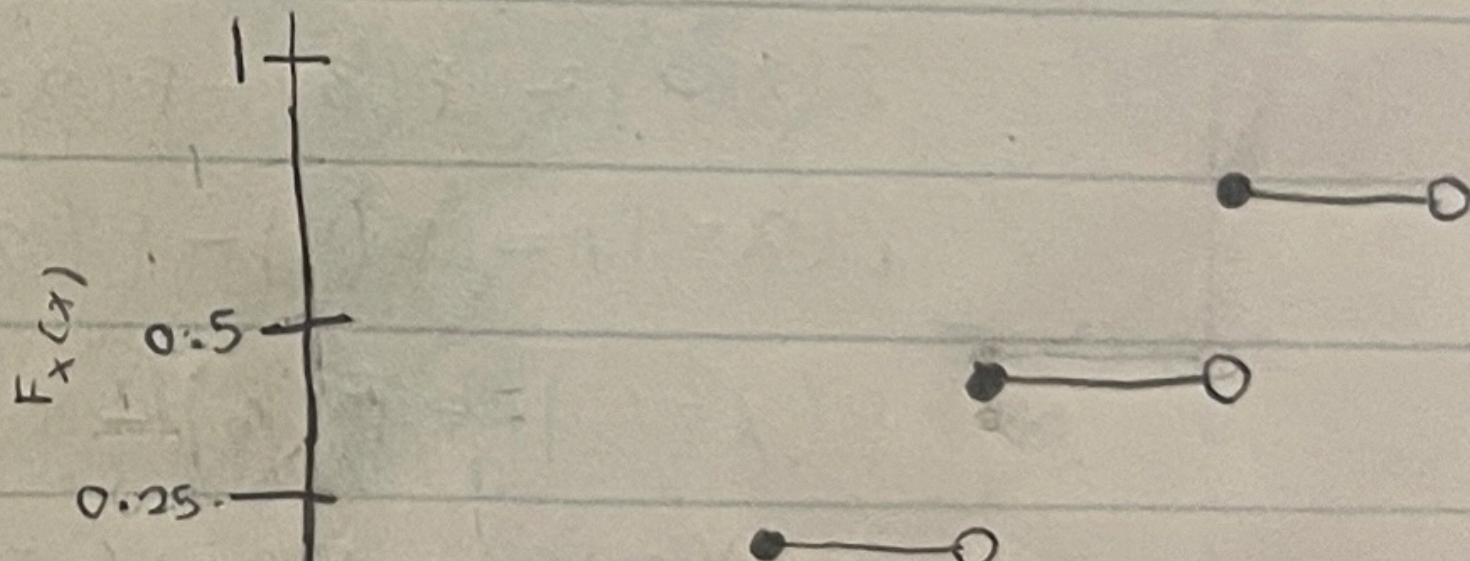
$P(X=3)$: outcomes $\Rightarrow (1,3), (3,1), (2,3), (3,2), (3,3) \rightarrow \frac{5}{36}$

$P(X=4)$: outcomes $\Rightarrow (1,4), (4,1), (2,4), (4,2), (3,4), (4,3), (4,4) \rightarrow \frac{7}{36}$

$P(X=5)$: outcomes $\Rightarrow (1,5), (5,1), (2,5), (5,2), (3,5), (5,3), (4,5), (5,4), (5,5) \rightarrow \frac{9}{36} = \frac{1}{4}$

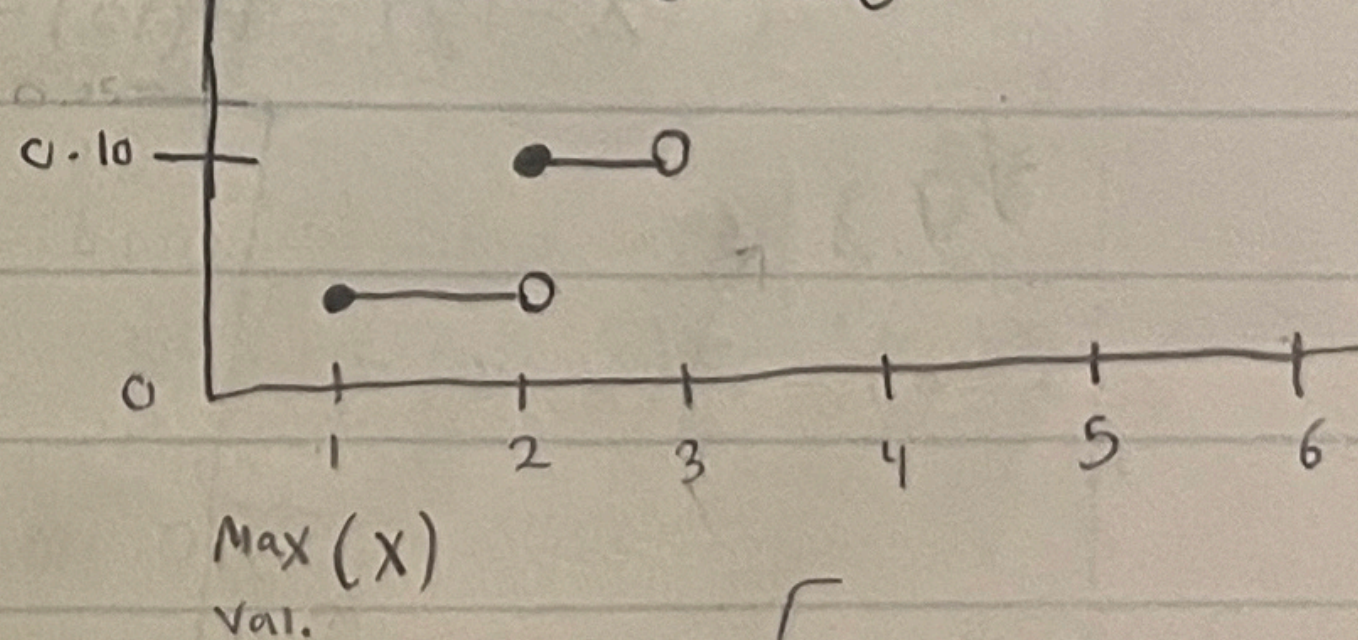
$P(X=6)$: outcomes $\Rightarrow (1,6), (6,1), (2,6), (6,2), (3,6), (6,3), (4,6), (6,4), (5,6), (6,5), (6,6) \rightarrow \frac{11}{36}$

$$P_X(x) = P(X=x) = \begin{cases} \frac{1}{36}, & x=1 \\ \frac{3}{36}, & x=2 \\ \frac{5}{36}, & x=3 \\ \frac{7}{36}, & x=4 \end{cases}$$



↑ CDF is prob.
that x is less
than or equal to a value

$$\left[\begin{array}{l} \frac{9}{36}, x=5 \\ \frac{11}{36}, x=6 \end{array} \right]$$



$$b) F_X(x) = P(X \leq x)$$

$$\hookrightarrow F(1) = P(X \leq 1) = \frac{1}{36}$$

$$F(2) = P(X \leq 2) = \frac{1}{36} + \frac{3}{36} = \frac{4}{36} = \frac{1}{9}$$

$$F(3) = P(X \leq 3) = \frac{1}{36} + \frac{3}{36} + \frac{5}{36} = \frac{9}{36} = \frac{1}{4}$$

$$F(4) = P(X \leq 4) = \frac{1}{36} + \frac{3}{36} + \frac{5}{36} + \frac{7}{36} = \frac{16}{36} = \frac{4}{9}$$

$$F(5) = P(X \leq 5) = \frac{1}{36} + \frac{3}{36} + \frac{5}{36} + \frac{7}{36} + \frac{9}{36} = \frac{25}{36}$$

$$F(6) = P(X \leq 6) = \frac{1}{36} + \frac{3}{36} + \frac{5}{36} + \frac{7}{36} + \frac{9}{36} + \frac{11}{36} = \frac{36}{36} = 1$$

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{36} & 1 \leq x < 2 \\ \frac{1}{9} & 2 \leq x < 3 \\ \frac{1}{4} & 3 \leq x < 4 \\ \frac{4}{9} & 4 \leq x < 5 \\ \frac{25}{36} & 5 \leq x < 6 \\ 1 & x \geq 6 \end{cases}$$

$F(x) =$

$$F(x) = \begin{cases} 0.30 & 1 \leq x < 3 \\ 0.40 & 3 \leq x < 4 \\ 0.45 & 4 \leq x < 6 \\ 0.60 & 6 \leq x < 12 \\ 1 & 12 \leq x \end{cases}$$

$$a) P(X=x) = F(x) - F(x^-)$$

$$\hookrightarrow P(X=1) = F(1) - F(x < 1) = 0.30 - 0 = 0.30$$

$$P(X=3) = F(3) - F(x < 3) = 0.40 - 0.30 = 0.10$$

$$P(X=4) = F(4) - F(x < 4) = 0.45 - 0.4 = 0.05$$

$$P(X=6) = F(6) - F(x < 6) = 0.60 - 0.45 = 0.15$$

$$P(X=12) = F(12) - F(x < 12) = 1 - 0.60 = 0.40$$

$$P(x) = P(X=x) = \begin{cases} 0.30 & x=1 \\ 0.10 & x=3 \\ 0.05 & x=4 \\ 0.15 & x=6 \\ & x=12 \end{cases} \rightarrow \text{PMF}$$

$$\begin{cases} 0.40 & x=12 \\ 0 & \text{otherwise} \end{cases}$$

$$b) P(3 \leq x \leq 6) = F(6) - F(3^-)$$

$$\hookrightarrow 0.60 - 0.30 = \boxed{0.30}$$

$$P(4 \leq x) = P(X \geq 4) = 1 - F(4^-) = 1 - 0.40 = \boxed{0.60}$$

$$\rightarrow \sum_{k=2}^{\infty} \frac{1}{3^k} = 1$$

$$\frac{1}{3^2} = \frac{1}{9}$$

geometric series

$$C \sum_{k=2}^{\infty} \frac{1}{3^k} = 1$$

$$\sum_{k=2}^{\infty} \frac{1}{3^k} = \frac{1/9}{1-1/3} = \frac{1/9}{2/3} = \frac{1}{6}$$

2, 4, 6, 8, ...

$$C \cdot \frac{1}{6} = 1 \Rightarrow \boxed{C=6}$$

can also leave
K as K and
account
for even
inputs (2, 4, 6, ...)

$$K=2n$$

$$n=1, 2, 3$$

$$* b) P(\text{outcome is even}) = \sum_{\substack{\text{even} \\ K}} P(K) = 6 \sum_{n=1}^{\infty} \frac{1}{3^{2n}}$$

$$= 6 \sum_{n=1}^{\infty} \frac{1}{3^{2n}} = 6 \sum_{n=1}^{\infty} \frac{1}{9^n} = 6 \cdot \frac{1/9}{1-1/9} = 6 \cdot \frac{1/9}{8/9} = \frac{6}{8} = \boxed{\frac{3}{4}}$$

c) P(outcome is less than or equal to 6)

$$\hookrightarrow P(K \leq 6) = \sum_{k=2}^6 P(K) = \sum_{k=2}^6 \frac{6}{3^k}$$

$$= P(2) + P(3) + P(4) + P(5) + P(6)$$

$$= \frac{6}{9} + \frac{6}{27} + \frac{6}{81} + \frac{6}{243} + \frac{6}{729}$$

$$= \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \frac{2}{81} + \frac{2}{243}$$

$$= \frac{102 + 54 + 18 + 6 + 2}{243} = \boxed{\frac{242}{243}}$$

Question assigned to the following page: [1](#)

$$P(A)(P(A)-1) = 0$$

· $P(A)=0, P(A)=1$ } $\rightarrow P(A)$ must be either 0
or 1.

